Lesson 46: LP Formulation and Occupancy Measures

AI/ML Mathematics

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1 Motivation and Overview

Lessons 42 and 45 developed two iterative algorithms for solving discounted MDPs: value iteration and policy iteration. Both are efficient and widely used, but they share a common limitation – they cannot easily accommodate *constraints*. Consider a robot navigation MDP where the agent must reach a goal while keeping the expected number of visits to dangerous states below a threshold, or an energy management system that must minimize cost subject to an average emission constraint. These constrained optimization problems cannot be directly formulated as standard Bellman equations.

Linear programming provides an alternative, fundamentally different approach to MDPs. Rather than iterating on value functions or policies, we cast the MDP as a linear program whose optimal solution *is* the optimal value function. The dual of this LP introduces *occupancy measures* – probability distributions over state-action pairs that encode how often each pair is visited under a policy. The one-to-one correspondence between stationary policies and occupancy measures transforms policy optimization into optimization over a convex polytope, enabling the incorporation of linear constraints naturally. This lesson develops the LP formulation, the duality theory, and the occupancy measure framework, culminating in the connection to constrained MDPs and inverse RL.

Why the LP Formulation Matters for RL/ML

- **Constrained MDPs**: Adding linear constraints on occupancy measures directly handles safety constraints, resource budgets, and fairness requirements – a key capability for real-world RL deployment.
- **Inverse RL**: The occupancy measure framework underpins inverse reinforcement learning, where the goal is to recover a reward function from observed expert behavior by matching occupancy measures.
- **Apprenticeship learning**: Altman [1999] and subsequent work on apprenticeship learning formulate imitation as matching the feature expectations (linear functions of occupancy measures) between the learner and the expert.
- **Convex optimization perspective**: The LP formulation reveals that MDP solving is a convex optimization problem, connecting RL theory to the rich toolkit of convex analysis (Lesson 17) and duality (Lesson 18).

Prerequisites. MDP model, policies, value functions (Lesson 39). Bellman optimality equations, the Bellman optimality operator $\mathcal{T}^*$ (Lesson 41). Value iteration (Lesson 42) and policy iteration (Lesson 45). Linear programming basics: primal/dual LPs, strong duality, basic feasible solutions, vertices of polytopes (Lesson 18 or equivalent background in convex optimization).

Learning objectives. After completing this lesson, the student will be able to:

- *Derive* the primal LP formulation of a discounted MDP and *prove* that its optimal solution equals V^*.
- *Define* occupancy measures and *prove* the flow conservation constraints they satisfy.
- *Establish* the one-to-one correspondence between stationary policies and occupancy measures.
- *Prove* LP strong duality for the MDP primal-dual pair.
- *Formulate* constrained MDPs as linear programs with additional constraints on occupancy measures.

- *Explain* how occupancy measures connect to inverse RL and apprenticeship learning.

2 References

- Puterman [2014, Chapter 6, Sections 6.9–6.10] – Definitive treatment of the LP formulation of discounted MDPs, occupancy measures, and duality.
- Manne [1960] – Original paper formulating MDPs as linear programs, establishing the connection between value functions and LP solutions.
- Altman [1999] – Comprehensive treatment of constrained MDPs, including the LP approach and Lagrangian relaxation methods.
- Bertsekas [2012, Chapter 1] – LP formulation in the context of dynamic programming and optimal control.
- Derman [1970] – Early treatment of the LP approach to MDPs with applications to operations research.
- Sutton and Barto [2018, Chapter 6] – Discussion of the relationship between value functions, policies, and optimization.

3 The Primal Linear Program

We work with a finite MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$ with $|\mathcal{S}| = S$, $|\mathcal{A}| = A$, and $\gamma \in [0, 1)$. The key insight is that the Bellman optimality equation $v(s) = \max_a[R(s, a) + \gamma \sum_{s'} P(s'|s, a)v(s')]$ can be relaxed into a system of linear inequalities.

Definition 3.1 (Bellman Feasibility)**.** A vector $v \in \mathbb{R}^S$ is *Bellman superharmonic* (or *Bellman feasible*) if

$$v(s) \geq R(s, a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s, a)\, v(s') \quad \text{for all } s \in \mathcal{S},\ a \in \mathcal{A}. \tag{1}$$

Lemma 3.2 (V^* Is the Smallest Bellman Superharmonic Vector)**.** *The optimal value function V^* satisfies the Bellman feasibility constraints (1). Moreover, if $v \in \mathbb{R}^S$ satisfies (1), then $v(s) \geq V^*(s)$ for all s.*

Proof. V^* satisfies (1) with equality at the maximizing action: $V^*(s) = \max_a[R(s, a)+\gamma \sum_{s'} P(s'|s, a)V^*(s')] \geq R(s, a) + \gamma \sum_{s'} P(s'|s, a)V^*(s')$ for all a.

Now suppose v satisfies (1). Then for all s:

$$v(s) \geq \max_{a \in \mathcal{A}} \left[R(s, a) + \gamma \sum_{s'} P(s'|s, a)\, v(s') \right] = (\mathcal{T}^* v)(s).$$

So $v \geq \mathcal{T}^* v$. By the monotonicity of $\mathcal{T}^*$ (Lesson 42, Proposition 8.1): $\mathcal{T}^* v \geq (\mathcal{T}^*)^2 v \geq \cdots$. Wait – this goes the wrong direction. Let us argue more carefully.

From $v \geq \mathcal{T}^* v$, by monotonicity of $\mathcal{T}^*$: $\mathcal{T}^* v \geq (\mathcal{T}^*)^2 v$. So $v \geq \mathcal{T}^* v \geq (\mathcal{T}^*)^2 v \geq \cdots$ The sequence $\{(\mathcal{T}^*)^n v\}$ is non-increasing. Since $(\mathcal{T}^*)^n v \to V^*$ by the Banach fixed-point theorem (Lesson 42, Theorem 4.1), we get $v \geq V^*$. □

Lemma 3.2 shows that V^* minimizes any objective that sums $v(s)$ with positive weights over the Bellman feasible set. This motivates the primal LP.

Definition 3.3 (Primal LP for Discounted MDPs)**.** Let $\alpha \in \mathbb{R}^S$ be any vector with $\alpha(s) > 0$ for all s and $\sum_s \alpha(s) = 1$ (an initial state distribution). The *primal LP* is:

$$\begin{aligned} \text{minimize} \quad & c^\top v = \sum_{s \in \mathcal{S}} \alpha(s)\, v(s) \\ \text{subject to} \quad & v(s) \geq R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)\, v(s'), \quad \forall\, s \in \mathcal{S},\ a \in \mathcal{A}, \\ & v \in \mathbb{R}^S. \end{aligned} \tag{2}$$

This LP has S decision variables and SA inequality constraints.

Theorem 3.4 (Primal LP Solves the MDP)**.** *The optimal value of the primal LP* (2) *equals* $\sum_s \alpha(s) V^*(s)$*, and the unique optimal solution is* $v^* = V^*$*.*

Proof. By Lemma 3.2, V^* is feasible and $v \geq V^*$ for every feasible v. Since $\alpha(s) > 0$ for all s:

$$\sum_s \alpha(s) v(s) \geq \sum_s \alpha(s) V^*(s)$$

for every feasible v, with equality if and only if $v = V^*$ (since $\alpha(s) > 0$ and $v(s) \geq V^*(s)$ imply that equality of the sums requires $v(s) = V^*(s)$ for all s). Therefore V^* is the unique minimizer. □

MDP as a Linear Program

Finding the optimal value function V^* of a discounted MDP is equivalent to solving the linear program (2). The LP has S variables and SA constraints. Once V^* is obtained, the optimal policy is recovered by identifying which constraints are active (tight) at each state: $\pi^*(s) = \arg\max_a [R(s,a) + \gamma \sum_{s'} P(s'|s,a) V^*(s')]$.

Remark 3.5. The choice of α does not affect the optimal *solution* $v^* = V^*$ (which is unique by Lemma 3.2), only the optimal *value* $\alpha^\top V^*$. Any strictly positive α works. The uniform distribution $\alpha(s) = 1/S$ is a common default.

Example 3.6 (Primal LP for a Two-State MDP)**.** Consider the two-state MDP with $\mathcal{S} = \{1, 2\}$, $\mathcal{A} = \{a_1, a_2\}$, $\gamma = 0.5$:

s	a	$R(s,a)$	$P(1 \mid s,a)$	$P(2 \mid s,a)$
1	a_1	5	0.5	0.5
1	a_2	10	0	1
2	a_1	-1	0.8	0.2
2	a_2	2	0.3	0.7

With $\alpha = (0.5, 0.5)^\top$, the primal LP is:

$$\begin{aligned} \min \quad & 0.5\, v_1 + 0.5\, v_2 \\ \text{s.t.} \quad & v_1 \geq 5 + 0.5(0.5\, v_1 + 0.5\, v_2) \\ & v_1 \geq 10 + 0.5(0 \cdot v_1 + 1 \cdot v_2) \\ & v_2 \geq -1 + 0.5(0.8\, v_1 + 0.2\, v_2) \\ & v_2 \geq 2 + 0.5(0.3\, v_1 + 0.7\, v_2) \end{aligned}$$

Simplifying:

$$\begin{aligned} 0.75\, v_1 - 0.25\, v_2 &\geq 5, \\ v_1 - 0.5\, v_2 &\geq 10, \\ -0.4\, v_1 + 0.9\, v_2 &\geq -1, \\ -0.15\, v_1 + 0.65\, v_2 &\geq 2. \end{aligned}$$

From Lessons 41–42, $V^* \approx (13.04, 6.09)^\top$, achieved by $\pi^*(1) = a_2$, $\pi^*(2) = a_2$. One can verify that V^* satisfies all four constraints, with the second and fourth constraints tight (corresponding to the optimal actions).

LP Size and Practical Considerations

The primal LP has S variables and SA constraints. For modern LP solvers, problems with up to 10^5–10^6 variables and constraints are tractable. However, for very large MDPs ($S > 10^6$), specialized algorithms (value iteration, policy iteration) are more practical. The LP formulation is most valuable for its theoretical insights (duality, occupancy measures, constrained MDPs) rather than as a computational workhorse.

4 The Dual LP and Occupancy Measures

We now derive the dual of the primal LP (2) and show that it has a beautiful probabilistic interpretation in terms of occupancy measures.

Theorem 4.1 (Dual LP for Discounted MDPs)**.** *The dual of the primal LP (2) is:*

$$\begin{aligned} &\textit{maximize} && \sum_{s,a} R(s,a)\, x(s,a) \\ &\textit{subject to} && \sum_{a \in \mathcal{A}} x(s,a) = \alpha(s) + \gamma \sum_{s',a'} P(s \mid s',a')\, x(s',a'), \quad \forall\, s \in \mathcal{S}, \\ & && x(s,a) \geq 0, \quad \forall\, s \in \mathcal{S},\ a \in \mathcal{A}. \end{aligned} \tag{3}$$

This LP has SA decision variables $x(s,a)$ and S equality constraints (plus SA non-negativity constraints).

Proof. Rewrite the primal constraints as: for all (s,a),

$$v(s) - \gamma \sum_{s'} P(s'|s,a) v(s') \geq R(s,a).$$

In matrix form with $v \in \mathbb{R}^S$: $(I - \gamma P_a^\top) v \geq r_a$ for each action a, where P_a is the $S \times S$ matrix with entries $P_a(s,s') = P(s'|s,a)\mathbf{1}[a]$. More precisely, we associate a dual variable $x(s,a) \geq 0$ with each constraint. The Lagrangian is:

$$\begin{aligned} L(v,x) &= \sum_s \alpha(s) v(s) - \sum_{s,a} x(s,a) \left[v(s) - R(s,a) - \gamma \sum_{s'} P(s'|s,a) v(s') \right] \\ &= \sum_s v(s) \left[\alpha(s) - \sum_a x(s,a) + \gamma \sum_{s',a'} P(s|s',a') x(s',a') \right] + \sum_{s,a} R(s,a) x(s,a). \end{aligned}$$

For the dual to be bounded ($\inf_v L(v,x) > -\infty$), the coefficient of each $v(s)$ must be zero:

$$\sum_a x(s,a) = \alpha(s) + \gamma \sum_{s',a'} P(s|s',a') x(s',a'), \quad \forall\, s.$$

Under this constraint, $L(v, x) = \sum_{s,a} R(s,a)x(s,a)$, which we maximize. This gives the dual LP (3). □

Definition 4.2 (Occupancy Measure)**.** For a stationary policy π and initial distribution α, the *(discounted) occupancy measure* is

$$x^\pi(s,a) = (1-\gamma)\sum_{t=0}^{\infty} \gamma^t \mathbb{P}_\alpha^\pi(S_t = s, A_t = a), \tag{4}$$

where $\mathbb{P}_\alpha^\pi$ denotes the probability under policy π with initial distribution $S_0 \sim \alpha$.

Remark 4.3. The normalization factor $(1-\gamma)$ ensures that x^π is a probability distribution: $\sum_{s,a} x^\pi(s,a) = 1$. Some references omit the $(1-\gamma)$ factor, in which case $\sum_{s,a} x^\pi(s,a) = 1/(1-\gamma)$. We adopt the normalized convention.

Without the normalization factor, define $\tilde{x}^\pi(s,a) = \sum_{t=0}^{\infty} \gamma^t \mathbb{P}_\alpha^\pi(S_t = s, A_t = a)$, so that $x^\pi = (1-\gamma)\tilde{x}^\pi$. We will show that $\tilde{x}^\pi$ is a feasible solution of the dual LP (3) (after appropriate rescaling).

Proposition 4.4 (Occupancy Measures as Dual Feasible Points)**.** *For any stationary policy π, the unnormalized occupancy measure $\tilde{x}^\pi(s,a) = \sum_{t=0}^{\infty} \gamma^t \mathbb{P}_\alpha^\pi(S_t = s, A_t = a)$ satisfies the dual constraints* (3)*.*

Proof. Clearly $\tilde{x}^\pi(s,a) \geq 0$. For the flow constraint, define the state occupancy $\tilde{\mu}^\pi(s) = \sum_a \tilde{x}^\pi(s,a) = \sum_{t=0}^{\infty} \gamma^t \mathbb{P}_\alpha^\pi(S_t = s)$. We need to show:

$$\tilde{\mu}^\pi(s) = \alpha(s) + \gamma \sum_{s',a'} P(s|s',a')\tilde{x}^\pi(s',a').$$

The left side is:

$$\begin{aligned}
\tilde{\mu}^\pi(s) &= \mathbb{P}_\alpha(S_0 = s) + \sum_{t=1}^{\infty} \gamma^t \mathbb{P}_\alpha^\pi(S_t = s) \\
&= \alpha(s) + \gamma \sum_{t=0}^{\infty} \gamma^t \sum_{s',a'} P(s|s',a')\mathbb{P}_\alpha^\pi(S_t = s', A_t = a') \\
&= \alpha(s) + \gamma \sum_{s',a'} P(s|s',a')\tilde{x}^\pi(s',a'),
\end{aligned}$$

where the second equality uses the Markov property: $\mathbb{P}_\alpha^\pi(S_{t+1} = s) = \sum_{s',a'} P(s|s',a')\mathbb{P}_\alpha^\pi(S_t = s', A_t = a')$. □

Proposition 4.5 (Dual Objective Equals Policy Value)**.** *The dual objective evaluated at $\tilde{x}^\pi$ equals the expected discounted return under π:*

$$\sum_{s,a} R(s,a)\tilde{x}^\pi(s,a) = \sum_s \alpha(s)V^\pi(s).$$

Proof.

$$\begin{aligned}\sum_{s,a} R(s,a)\tilde{x}^\pi(s,a) &= \sum_{s,a} R(s,a) \sum_{t=0}^{\infty} \gamma^t \mathbb{P}_\alpha^\pi(S_t = s, A_t = a) \\ &= \sum_{t=0}^{\infty} \gamma^t \sum_{s,a} R(s,a)\mathbb{P}_\alpha^\pi(S_t = s, A_t = a) \\ &= \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_\alpha^\pi[R(S_t, A_t)] \\ &= \mathbb{E}_\alpha^\pi\left[\sum_{t=0}^{\infty} \gamma^t R(S_t, A_t)\right] = \sum_s \alpha(s) V^\pi(s).\end{aligned}$$

□

5 Properties of Occupancy Measures

Theorem 5.1 (Characterization of the Occupancy Polytope)**.** *Define the* occupancy polytope

$$\mathcal{X}_\alpha = \left\{ x \in \mathbb{R}_{\geq 0}^{SA} : \sum_a x(s,a) = \alpha(s) + \gamma \sum_{s',a'} P(s|s',a')x(s',a'), \ \ \forall\, s \in \mathcal{S} \right\}. \tag{5}$$

Then:

1. *$\mathcal{X}_\alpha$ is a non-empty bounded convex polytope.*
2. *$\sum_{s,a} x(s,a) = 1/(1-\gamma)$ for every $x \in \mathcal{X}_\alpha$.*
3. *The map $\pi \mapsto \tilde{x}^\pi$ is a bijection between stationary policies and $\mathcal{X}_\alpha$.*

Proof. **Part 1.** Non-emptiness: $\tilde{x}^\pi \in \mathcal{X}_\alpha$ for any stationary policy π (Proposition 4.4). Convexity: $\mathcal{X}_\alpha$ is defined by linear equalities and non-negativity constraints, hence it is a polyhedron. Boundedness: summing the flow constraints over s:

$$\sum_{s,a} x(s,a) = 1 + \gamma \sum_{s,a} x(s,a),$$

so $\sum_{s,a} x(s,a) = 1/(1-\gamma)$. This proves Part 2 and also implies boundedness (each $x(s,a) \leq 1/(1-\gamma)$).

Part 3 (surjectivity). Given $x \in \mathcal{X}_\alpha$, define $\mu(s) = \sum_a x(s,a)$ and the policy

$$\pi_x(a|s) = \begin{cases} x(s,a)/\mu(s) & \text{if } \mu(s) > 0, \\ 1/|\mathcal{A}| & \text{if } \mu(s) = 0. \end{cases} \tag{6}$$

We need to show that $\tilde{x}^{\pi_x} = x$. Let $y = \tilde{x}^{\pi_x}$. Define $\nu(s) = \sum_a y(s,a)$. Then $y(s,a) = \pi_x(a|s)\nu(s)$ and $\nu(s) = \alpha(s) + \gamma \sum_{s'} P(s|s', \pi_x(s'))\nu(s')$. Similarly, $x(s,a) = \pi_x(a|s)\mu(s)$ by construction and $\mu(s) = \alpha(s) + \gamma \sum_{s',a'} P(s|s',a')x(s',a') = \alpha(s) + \gamma \sum_{s'} P(s|s', \pi_x(s'))\mu(s')$. So both μ and ν satisfy the same linear system $\mu = \alpha + \gamma(P^{\pi_x})^\top \mu$, i.e., $(I - \gamma(P^{\pi_x})^\top)\mu = \alpha$. Since $\gamma < 1$, this system has a unique solution (Lesson 45, Theorem **??** applies to the transpose), so $\mu = \nu$ and hence $x = y$.

Part 3 (injectivity). If $\tilde{x}^{\pi_1} = \tilde{x}^{\pi_2}$, then by (6), $\pi_1(a|s) = \tilde{x}^{\pi_1}(s,a)/\mu(s) = \tilde{x}^{\pi_2}(s,a)/\mu(s) = \pi_2(a|s)$ for all s with $\mu(s) > 0$. For states with $\mu(s) = 0$ (never visited), the policies may differ, but they yield the same occupancy measure. □

Policy-Occupancy Correspondence

There is a one-to-one correspondence between stationary policies and points in the occupancy polytope $\mathcal{X}_\alpha$:

$$\pi \longleftrightarrow \tilde{x}^\pi, \qquad \pi_x(a|s) = \frac{x(s,a)}{\sum_{a'} x(s,a')}.$$

Deterministic policies correspond to vertices of $\mathcal{X}_\alpha$, and the optimal policy corresponds to the vertex that maximizes the linear objective $\sum_{s,a} R(s,a)x(s,a)$.

Proposition 5.2 (Deterministic Policies Are Vertices)**.** *The vertices (extreme points) of $\mathcal{X}_\alpha$ are precisely the occupancy measures of deterministic stationary policies.*

Proof. $\mathcal{X}_\alpha$ is a polytope in $\mathbb{R}^{SA}$ defined by S equality constraints and SA non-negativity constraints. A vertex (basic feasible solution) has exactly $SA - S = S(A-1)$ active non-negativity constraints, meaning that for each state s, exactly $A-1$ of the variables $\{x(s,a)\}_{a\in\mathcal{A}}$ are zero. Hence exactly one action per state has $x(s,a) > 0$, which corresponds to a deterministic policy.

Conversely, for a deterministic policy d, the occupancy measure $\tilde{x}^d(s,a) = 0$ for $a \neq d(s)$, giving exactly $S(A-1)$ active non-negativity constraints. The remaining S variables $\{\tilde{x}^d(s,d(s))\}_{s\in\mathcal{S}}$ are uniquely determined by the S flow constraints (since the linear system is non-singular). Thus $\tilde{x}^d$ is a basic feasible solution, hence a vertex. □

Example 5.3 (Occupancy Measures for the Two-State MDP)**.** Consider the two-state MDP from Example 3.6 with $\gamma = 0.5$ and $\alpha = (0.5, 0.5)^\top$.

For the optimal policy $d^*(1) = a_2$, $d^*(2) = a_2$, the transition matrix is $P^{d^*} = \left(\begin{smallmatrix} 0 & 1 \\ 0.3 & 0.7 \end{smallmatrix}\right)$. The unnormalized state occupancy satisfies $(I - 0.5(P^{d^*})^\top)\mu = \alpha$:

$$\begin{pmatrix} 1 & -0.15 \\ -0.5 & 0.65 \end{pmatrix} \mu = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}.$$

Determinant $= 0.65 - 0.075 = 0.575$. $\mu_1 = (0.5 \cdot 0.65 + 0.15 \cdot 0.5)/0.575 = 0.4/0.575 \approx 0.696$, $\mu_2 = (1 \cdot 0.5 + 0.5 \cdot 0.5)/0.575 = 0.75/0.575 \approx 1.304$. Check: $\mu_1 + \mu_2 = 2 = 1/(1-0.5)$. Correct.

The occupancy measure is $\tilde{x}^{d^*}(1, a_2) = 0.696$, $\tilde{x}^{d^*}(2, a_2) = 1.304$, and $\tilde{x}^{d^*}(s, a_1) = 0$ for both states (deterministic policy).

The dual objective is $10 \cdot 0.696 + 2 \cdot 1.304 = 9.57$, which should equal $\alpha^\top V^* = 0.5 \cdot 13.04 + 0.5 \cdot 6.09 = 9.57$. Confirmed.

The normalized occupancy is $x^{d^*} = (1-0.5)\tilde{x}^{d^*}$: $x^{d^*}(1, a_2) = 0.348$, $x^{d^*}(2, a_2) = 0.652$. This is a probability distribution summing to 1, representing the long-run discounted frequency of visiting each state-action pair.

6 LP Strong Duality

Theorem 6.1 (Strong Duality for the MDP LP)**.** *The primal LP (2) and dual LP (3) satisfy strong duality:*

$$\min_{v \text{ feasible}} \sum_s \alpha(s)v(s) = \max_{x\in\mathcal{X}_\alpha} \sum_{s,a} R(s,a)x(s,a) = \sum_s \alpha(s)V^*(s). \tag{7}$$

The unique primal optimum is $v^* = V^*$. The dual optimum is $x^* = \tilde{x}^{\pi^*}$, the unnormalized occupancy measure of any optimal policy π^*.

Proof. Both the primal and dual are feasible: the primal is feasible at V^* and the dual is feasible at $\tilde{x}^\pi$ for any π. Both are bounded: the primal is bounded below (by the dual objective at any feasible x) and the dual is bounded above (by the primal objective at V^*). By the LP strong

duality theorem (which applies to any LP pair where both are feasible and bounded), the optimal values are equal.

By Theorem 3.4, the primal optimum is V^*. By Proposition 4.5, the dual objective at $\tilde{x}^{\pi^*}$ equals $\sum_s \alpha(s)V^*(s)$, which is the primal optimal value. Hence $\tilde{x}^{\pi^*}$ is dual optimal. □

Remark 6.2. Complementary slackness gives additional insight. At the optimal pair $(V^*, \tilde{x}^{\pi^*})$:

- For each (s, a): either $\tilde{x}^{\pi^*}(s, a) = 0$ or the constraint $V^*(s) = R(s, a)+\gamma \sum_{s'} P(s'|s, a)V^*(s')$ is tight.
- This means the optimal policy only assigns positive probability to actions that are optimal (achieve the maximum in the Bellman equation) – consistent with the greedy policy characterization.

LP Duality and Reward Shaping

The LP duality framework provides a principled way to understand reward shaping. A potential-based shaping function $\Phi : \mathcal{S} \to \mathbb{R}$ modifies the reward to $\tilde{R}(s, a, s') = R(s, a) + \gamma\Phi(s') - \Phi(s)$. In the LP framework, this corresponds to translating the primal variable $v \mapsto v - \Phi$, which shifts the constraints but does not change the optimal policy. The occupancy measures (dual variables) remain unchanged – confirming that potential-based shaping preserves the optimal policy, a result known as Ng's reward shaping theorem.

7 Constrained MDPs

The LP formulation naturally accommodates constraints on the agent's behavior, leading to constrained MDPs (CMDPs).

Definition 7.1 (Constrained MDP)**.** A *constrained MDP* (CMDP) extends a standard MDP with K auxiliary cost functions $C_k : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ $(k = 1, \dots, K)$ and corresponding bounds b_k. The goal is:

$$\max_{\pi} \sum_s \alpha(s)V^\pi(s) \quad \text{subject to} \quad \sum_s \alpha(s)V^\pi_{C_k}(s) \le b_k, \quad k = 1, \dots, K, \tag{8}$$

where $V^\pi_{C_k}(s) = \mathbb{E}^\pi[\sum_{t=0}^\infty \gamma^t C_k(S_t, A_t) \mid S_0 = s]$ is the expected discounted cost under cost function C_k.

The key observation is that $\sum_s \alpha(s)V^\pi_{C_k}(s) = \sum_{s,a} C_k(s, a)\tilde{x}^\pi(s, a)$ by the same argument as Proposition 4.5. Therefore, the CMDP can be written entirely in terms of occupancy measures.

Theorem 7.2 (LP Formulation of CMDPs)**.** *The constrained MDP* (8) *is equivalent to the linear program:*

$$\begin{aligned}
&\textit{maximize} && \sum_{s,a} R(s, a)\, x(s, a) \\
&\textit{subject to} && \sum_a x(s, a) = \alpha(s) + \gamma \sum_{s',a'} P(s|s', a')x(s', a'), \quad \forall\, s, \\
& && \sum_{s,a} C_k(s, a)\, x(s, a) \le b_k, \quad k = 1, \dots, K, \\
& && x(s, a) \ge 0, \quad \forall\, s, a.
\end{aligned} \tag{9}$$

Proof. By the policy-occupancy correspondence (Theorem 5.1), optimizing over stationary policies is equivalent to optimizing over $\mathcal{X}_\alpha$. The objective $\sum_s \alpha(s)V^\pi(s) = \sum_{s,a} R(s, a)\tilde{x}^\pi(s, a)$ and the constraints $\sum_s \alpha(s)V^\pi_{C_k}(s) = \sum_{s,a} C_k(s, a)\tilde{x}^\pi(s, a) \le b_k$ are all linear in $\tilde{x}^\pi$. Replacing $\tilde{x}^\pi$ by the free variable $x \in \mathcal{X}_\alpha$ yields (9). □

Constrained MDPs via Linear Programming

Any finite constrained MDP with K linear cost constraints can be solved as a linear program with SA variables and $S+K$ equality/ inequality constraints (plus non-negativity). The optimal policy may be *stochastic* – it randomizes over at most $K+1$ actions at each state (since the LP optimum can lie on a face of dimension at most K above the unconstrained vertex).

Remark 7.3. In the unconstrained case ($K=0$), the LP optimum is a vertex of $\mathcal{X}_\alpha$, corresponding to a deterministic policy. With K constraints, the optimum may lie in the interior of a face, requiring a mixture of at most $K+1$ deterministic policies. This is a fundamental result: constraints can force optimal policies to be stochastic.

Example 7.4 (Constrained Two-State MDP)**.** Consider the two-state MDP from Example 3.6 with an additional safety cost $C(s,a)$ representing the probability of triggering an alarm:

s	a	$C(s,a)$
1	a_1	0.1
1	a_2	0.5
2	a_1	0.2
2	a_2	0.3

We require the expected discounted alarm cost to be at most $b=0.6$: $\sum_{s,a} C(s,a)\tilde{x}(s,a) \le 0.6$.

The unconstrained optimum uses $d^*(1)=a_2$, $d^*(2)=a_2$ with $\tilde{x}^{d^*}(1,a_2)\approx 0.696$ and $\tilde{x}^{d^*}(2,a_2)\approx 1.304$. The alarm cost is $0.5\cdot 0.696+0.3\cdot 1.304 = 0.348+0.391 = 0.739 > 0.6$. The constraint is violated, so the unconstrained optimal policy is infeasible.

The constrained LP adds the constraint $0.1\,x(1,a_1)+0.5\,x(1,a_2)+0.2\,x(2,a_1)+0.3\,x(2,a_2) \le 0.6$. Solving this modified LP yields a stochastic policy that randomizes between a_1 and a_2 at one or both states.

Constrained MDPs in Safe RL

The CMDP framework is the theoretical foundation for *safe RL*, where agents must satisfy safety or resource constraints:

- **Constrained Policy Optimization (CPO)** solves an approximate CMDP at each iteration using trust-region methods combined with Lagrangian relaxation.
- **Lagrangian relaxation**: The CMDP can be relaxed by introducing multipliers $\lambda_k \ge 0$ for each constraint, yielding the unconstrained problem $\max_\pi[\sum_s \alpha(s)V^\pi(s) - \sum_k \lambda_k(\sum_s \alpha(s)V^\pi_{C_k}(s) - b_k)]$. Under strong duality (which holds for CMDPs), the saddle-point problem $\max_\pi \min_{\lambda\ge 0}$ yields the CMDP solution.
- **Primal-dual algorithms**: Practical safe RL algorithms alternate between policy updates and dual variable updates, implementing a stochastic primal-dual method on the CMDP Lagrangian.

8 Connection to Inverse RL and Apprenticeship Learning

The occupancy measure framework provides a natural bridge to *inverse reinforcement learning* (IRL), where the goal is to learn a reward function from observed expert behavior.

Definition 8.1 (Feature Expectations)**.** Given a feature map $\phi : \mathcal{S} \times \mathcal{A} \to \mathbb{R}^d$, the *feature expectation* of a policy π is

$$\bar{\phi}^\pi = \sum_{s,a} \phi(s,a)\, \tilde{x}^\pi(s,a) = \mathbb{E}^\pi_\alpha \left[\sum_{t=0}^{\infty} \gamma^t \phi(S_t, A_t) \right]. \tag{10}$$

Proposition 8.2 (Linear Reward and Feature Expectations)**.** *If the reward function is linear in features, $R(s,a) = w^\top \phi(s,a)$ for some weight vector $w \in \mathbb{R}^d$, then*

$$\sum_s \alpha(s) V^\pi(s) = w^\top \bar{\phi}^\pi.$$

Proof. Direct computation: $\sum_s \alpha(s)V^\pi(s) = \sum_{s,a} R(s,a)\tilde{x}^\pi(s,a) = \sum_{s,a} w^\top \phi(s,a)\tilde{x}^\pi(s,a) = w^\top \bar{\phi}^\pi$. □

Remark 8.3. In inverse RL, we observe an expert policy π_E (through demonstrations) and wish to find a reward function R under which π_E is optimal. If $R(s,a) = w^\top \phi(s,a)$, then π_E is optimal iff $w^\top \bar{\phi}^{\pi_E} \geq w^\top \bar{\phi}^\pi$ for all π. Equivalently, the expert's feature expectation $\bar{\phi}^{\pi_E}$ must lie on the boundary of the set $\{\bar{\phi}^\pi : \pi \text{ stationary}\}$ in the direction w.

This geometric view leads to algorithms that find w by searching for a hyperplane separating $\bar{\phi}^{\pi_E}$ from the feature expectations of other policies – the basis of *apprenticeship learning* and *maximum margin IRL*.

Example 8.4 (Feature Matching in IRL)**.** Consider a grid-world MDP with features $\phi(s,a) = (\mathbf{1}[s \text{ is near obstacle}], \mathbf{1}[s \text{ is at goal}], 1)^\top$. An expert demonstrates trajectories, from which we estimate $\bar{\phi}^{\pi_E} \approx (0.1, 0.8, 5.0)^\top$.

An apprenticeship learning algorithm seeks a policy π whose feature expectations $\bar{\phi}^\pi$ are close to $\bar{\phi}^{\pi_E}$. In the occupancy measure framework, this is:

$$\min_{x \in \mathcal{X}_\alpha} \left\| \sum_{s,a} \phi(s,a) x(s,a) - \bar{\phi}^{\pi_E} \right\|_2^2.$$

This is a quadratic program over the occupancy polytope, which is tractable and yields a policy that approximately matches the expert's behavior.

9 Exercises

Exercise 9.1 (Primal LP Construction)**.** $(*)$
Consider the MDP with $\mathcal{S} = \{1,2,3\}$, $\mathcal{A} = \{a,b\}$, $\gamma = 0.9$:

s	a	$R(s,a)$	$P(1\|s,a)$	$P(2\|s,a)$	$P(3\|s,a)$
1	a	2	0.5	0.3	0.2
1	b	0	0	0.5	0.5
2	a	1	0.4	0.4	0.2
2	b	3	0	0	1
3	a	-1	0.1	0.1	0.8
3	b	0	0	0.5	0.5

(a) Write out the primal LP with $\alpha = (1/3, 1/3, 1/3)$.

(b) For each deterministic policy, compute its value function and verify that the optimal one solves the LP.

Exercise 9.2 (Occupancy Measure Computation). $(*)$
For the two-state MDP in Example 3.6 with $\alpha = (0.5, 0.5)$:

(a) Compute the unnormalized occupancy measure $\tilde{x}^d$ for the policy $d(1) = a_1$, $d(2) = a_2$.

(b) Verify the flow conservation constraints.

(c) Compute the dual objective and compare with $\alpha^\top V^d$.

Exercise 9.3 (Vertices of the Occupancy Polytope). $(**)$
For a two-state, two-action MDP with $\gamma = 0.5$ and $\alpha = (0.5, 0.5)$:

(a) How many vertices does $\mathcal{X}_\alpha$ have?

(b) Write out the four flow conservation constraints (two equalities, four non-negativity constraints).

(c) Enumerate all vertices by solving the linear system with each combination of two zero variables.

(d) Identify which vertex corresponds to which deterministic policy.

Exercise 9.4 (CMDP with a Budget Constraint). $(**)$
Consider the MDP from Exercise 1 with an additional fuel cost $C(s, a)$: $C(1, a) = 1$, $C(1, b) = 3$, $C(2, a) = 2$, $C(2, b) = 1$, $C(3, a) = 0$, $C(3, b) = 2$. The budget is $b = 3$.

(a) Write the CMDP as an LP.

(b) Check whether the unconstrained optimal policy satisfies the fuel constraint.

(c) If not, describe qualitatively how the constrained optimal policy differs from the unconstrained one.

Exercise 9.5 (Stochastic Policies from CMDPs). $(***)$
Prove that for a CMDP with K cost constraints, the optimal policy randomizes over at most $K + 1$ actions at each state.

Hint: Use the fact that the optimal solution of an LP lies on a face of the feasible polytope, and count the number of additional constraints (beyond the original SA non-negativity constraints) that can become active.

Exercise 9.6 (Feature Expectations and IRL). $(***)$
Consider a three-state, two-action MDP with features $\phi(s, a) \in \mathbb{R}^2$. An expert policy π_E has feature expectation $\bar{\phi}^{\pi_E} = (3.0, 1.5)^\top$.

(a) Show that $\bar{\phi}^\pi$ is a linear function of the occupancy measure $\tilde{x}^\pi$.

(b) Prove that the set $\mathcal{F} = \{\bar{\phi}^\pi : \pi \text{ stationary}\}$ is a convex polytope whose vertices correspond to deterministic policies.

(c) Explain geometrically why there always exists a weight vector w such that π_E is optimal for the reward $R(s, a) = w^\top \phi(s, a)$, provided $\bar{\phi}^{\pi_E}$ lies on the boundary of $\mathcal{F}$.

10 Summary

- The *primal LP* formulation casts finding V^* as minimizing $\alpha^\top v$ subject to the Bellman feasibility constraints $v(s) \geq R(s,a) + \gamma \sum_{s'} P(s'|s,a)v(s')$ for all (s,a). The unique optimal solution is V^*.
- The *dual LP* introduces occupancy measure variables $x(s,a) \geq 0$ and maximizes $\sum_{s,a} R(s,a)x(s,a)$ subject to flow conservation constraints. The optimal dual solution is the occupancy measure of the optimal policy.
- *Occupancy measures* $x^\pi(s,a) = (1-\gamma)\sum_t \gamma^t \mathbb{P}^\pi(S_t = s, A_t = a)$ encode the discounted visitation frequencies. They satisfy flow conservation, sum to 1 (when normalized), and are non-negative.
- There is a *one-to-one correspondence* between stationary policies and points in the occupancy polytope: $\pi(a|s) = x(s,a)/\sum_{a'} x(s,a')$. Deterministic policies correspond to vertices.
- *LP strong duality* holds: the primal and dual optimal values are equal. Complementary slackness confirms that optimal occupancy measures concentrate on optimal actions.
- *Constrained MDPs* are naturally handled by adding linear constraints $\sum_{s,a} C_k(s,a)x(s,a) \leq b_k$ to the dual LP. The optimal constrained policy may be stochastic, randomizing over at most $K+1$ actions.
- *Feature expectations* $\bar{\phi}^\pi = \sum_{s,a} \phi(s,a)\tilde{x}^\pi(s,a)$ provide the foundation for inverse RL and apprenticeship learning, where policy optimization is reduced to matching feature statistics.

Since this is the final lesson in the MDP subdomain, we provide a summary of all eight MDP lessons.

Lesson	Title	Key Results
39	MDP Model and Policies	MDP tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$; policy classes (Markov, stationary, deterministic); value functions V^π, Q^π; Bellman expectation equations
40	Finite-Horizon MDPs	Backward induction; time-dependent optimal policy; dynamic programming principle
41	Discounted Reward and Bellman Equations	Bellman operators $\mathcal{T}^\pi$, $\mathcal{T}^*$; γ-contraction; unique fixed points V^π, V^*; policy improvement theorem; greedy optimality
42	Contraction and Value Iteration	VI algorithm $v^{(n+1)} = \mathcal{T}^* v^{(n)}$; geometric convergence at rate γ; a priori/a posteriori bounds; approximate policy extraction
43	VI Error Bounds and Monotonicity	Span seminorm bounds; monotone VI sequences; improved stopping criteria; early policy convergence
44	Average Reward MDPs	Gain-bias equations; unichain theory; relative value iteration; $\gamma \to 1$ limit; Blackwell optimality
45	Policy Iteration	Policy evaluation ($O(S^3)$ solve); policy improvement theorem; Howard's improvement; finite convergence ($\leq A^S$); modified PI; VI vs. PI comparison
46	LP Formulation and Occupancy Measures	Primal/dual LP; occupancy measures; policy-occupancy correspondence; strong duality; constrained MDPs; inverse RL connection

11 Further Reading

- **Lesson 24 (TD Learning)** extends the MDP theory developed in Lessons 39–46 to the model-free setting, where the transition kernel P and reward function R are unknown.
- **Lesson 25 (Policy Gradient)** develops policy optimization methods that parameterize the policy directly, using gradient ascent on the expected return. The occupancy measure perspective provides an alternative derivation of the policy gradient theorem.
- Puterman [2014, Chapters 6 and 8–9] provides the definitive treatment of LP formulations for both discounted and average reward MDPs, including multichain theory and sensitivity analysis.
- Manne [1960] is the original paper establishing the LP formulation; it is concise and elegant.
- Altman [1999] is the standard reference for constrained MDPs, covering LP methods, Lagrangian duality, sensitivity analysis, and applications.
- Derman [1970] provides an early treatment with applications to inventory and queuing problems.

- Ng and Russell (2000) introduced inverse RL from the occupancy measure perspective; Abbeel and Ng (2004) developed apprenticeship learning via feature expectation matching. These works directly build on the LP/occupancy framework of this lesson.
- For connections to convex optimization, see the treatment of LP duality in Lesson 18 (Convex Optimization and Duality).

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